

Appendix D: Dataset Description

This appendix documents the complete procedure used to generate, label, pre-process and partition the dataset that supports the XGBoost-based Q-scheduling model. All values reported below are taken from the data-generation script (MATLAB / MATPOWER) and the machine-learning script (Python), so that the study can be reproduced. The steady-state security of the system is checked with the MATPOWER power flow (runpf) followed by the limit-checking routine checklimits, which returns the sets of violated bus voltages, generator active / reactive outputs and branch flow limits.

D.1 Input Variables and Feature Definitions

Each sample is one feasible pre-contingency operating point. The input vector is the concatenation of three physically distinct blocks, listed in order of appearance in the data matrix:

Table D.1 Input feature specification.

Block	Definition	Dimension
1	Active power generation from synchronous generators	Number of synchronous generators
2	Active power generation from renewable generators	Number of renewable generators
3	Active power demand at a load bus after independent regulation	Number of local power loads

D.2 Table D.2 Value ranges of the generated variables.

Quantity	Range / value	Remark
Renewable active output	$0.90 \sim 1.10 \times \text{base value}$	Independent uniform draw per unit
Renewable reactive output	$0.90 \sim 1.10 \times \text{base value}$	Scaled with the active output of the same unit
Load active demand	$0.95 \sim 1.05 \times \text{base value}$	Independent uniform draw per bus
Load reactive demand	$0.95 \sim 1.05 \times \text{base value}$	Scaled with the active demand of the same bus
Thermal active output	$[P_{\min}, P_{\max}]$ of each unit	--
Target P_d (label)	$0 \sim 200 \times 10^4 \text{ kW}$	Upper bound P_{d0} , resolution $tol = 1 \times 10^4 \text{ kW}$

D.2 Output Variable and Dataset Generation

The target variable is the largest DC export power of the sending-end converter station for which the system remains secure both before and after an HVDC blocking event. For each candidate operating point the search starts from the rated value $P_d = P_{d0} = 200$ and decreases P_d in steps of $tol = 0.5\% P_{d0}$. At each trial value two AC power flows are solved:

- **Pre-contingency check:** the converter station is modelled as an additional load at the sending bus together with the net reactive exchange Q_c , and the resulting operating point must be feasible (power flow converged and no limit violation).
- **Post-blocking check:** the DC export is set to zero by injecting $P_{d1} = -P_d$ at the sending bus (with the corresponding reactive change $Q_{d1} = P_{d1} \tan \phi$), the resulting operating point must again be feasible.

The first trial value of P_d that passes both checks is recorded as the label $y = P_d$, and the sample is accepted. If the power flow diverges or any limit is violated, P_d is reduced by one step and the two checks are repeated; the loop terminates when $P_d < 0$, in which case no secure export level exists and the operating point is discarded without being written to the dataset.